

VISUALIZING GEODESICS IN SCHWARSCHILD METRIC

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Project Overview

UNVEILING GRAVITY'S INFLUENCE

This project delves into the motion of objects within the curved spacetime around a massive body, utilizing the Schwarzschild solution in general relativity. We establish the mathematical groundwork, derive geodesic equations, and solve them numerically using Python.

The ultimate goal is to visually comprehend how gravity shapes motion in curved spacetime through simulations.

Differential Geometry

General relativity describes gravity as a consequence of spacetime geometry. Differential geometry provides the mathematical tools to understand this.

Key Concepts

Topological Spaces:	Defining continuity and neighbourhoods.
Differential Manifolds:	Spaces locally homeomorphic to $\mathbb{R}^n$, allowing calculus.
Tangent Spaces:	Differentiating functions at a point, forming vector spaces.
Vector Fields:	Smooth assignment of tangent vectors, generating integral curves.
Lie Derivative:	Differentiating vector fields by comparing vectors from different spaces.
Covectors & One-Forms:	Dual spaces to tangent vectors, mapping vector fields to real numbers.
Tensors & Tensor Fields:	Multi-linear maps capturing complex geometric properties.
Differential Forms:	Anti-symmetric tensor fields used for integration.

Riemannian and Lorentzian Geometry

These geometries define how distances, angles and volumes are measured in the curved spaces.

The Metric:	A symmetric, non-degenerate $(0, 2)$ tensor field defining distances and angles.
Riemannian Manifolds:	Spaces with positive-definite metrics, like Euclidean space.
Lorentzian Manifolds:	Spaces with metrics like Minkowski space, crucial for general relativity.
Connection:	A map defining covariant derivatives, generalising partial derivatives to curved space.
Torsion & Riemann Tensor:	Tensors measuring the non-commutativity of covariant derivatives and spacetime curvature.
Levi-Civita Connection:	A unique torsion-free connection compatible with the metric.
Parallel Transport:	Connecting tangent spaces at different points, showing how tensors move along a path.
Geodesics:	Curves representing the "straightest" possible paths in curved spacetime, along which particles move freely.

The Geodesic Equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\rho\nu} \frac{dx^\rho}{d\tau} \frac{dx^\nu}{d\tau} = 0$$

x^μ Coordinates representing the position of the test particle

$\Gamma^\mu_{\rho\nu}$ Christoffel Symbols

τ Proper time parametrising the curve

The solutions of this equation are known as geodesics. These curves represent the paths which free particles follow in that specified geometry.

General Theory of Relativity

Equivalence principle

Gravitation is indistinguishable from acceleration, and can be locally transformed away.

Gravity as geometry

Accelerated motion in natural coordinates leads to a modified metric, implying gravity manifests as spacetime metric modification.

Einstein Field Equations

Equations governing the dynamics of the gravitational field, relating spacetime curvature to matter and energy.

Schwarzschild Metric

A solution to Einstein's equations describing the gravitational field of a non-rotating black hole.

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Python Program: Numerical Integration

To simulate geodesics, we need to solve the geodesic equation. We implement this using `scipy.integrate.solve_ivp`. First, we define the Christoffel symbols.

```
def trt(r): return M/(r*(r-2*M))
def ttr(r): return trt(r)
def rtt(r): return M*(r-2*M)/(r*r*r)
def rrr(r): return (-1)*M/(r*(r-2*M))
def rthth(r): return (-1)*(r-2*M)
def rpp(r, th): return (-1)*(r-2*M)*np.square(np.sin(th))
def thrth(r): return 1/r
def ththr(r): return 1/r
def thpp(th): return (-1)*np.sin(th)*np.cos(th)
def prp(r): return 1/r
def ppr(r): return 1/r
def pthp(th): return 1/np.tan(th)
def ppth(th): return pthp(th)
```


Python Program: Numerical Integration

We create a function which defines the state vector and the differential equations to be solved

```
def system(tau, vars):  
    t, r, th, p, ut, ur, uth, up = vars  
    dt = ut  
    dr = ur  
    dth = uth  
    dp = up  
    dut = 2*(-1)*trt(r)*ur*ut  
    dur = -rtt(r)*ut*ut - rrr(r)*ur*ur - rthth(r)*uth*uth - rthth(r)*up*up  
    duth = -2*thrth(r)*ur*uth - thpp(th)*up*up  
    dup = -2*prp(r)*ur*up - 2*pthp(th)*uth*up  
    return [dt, dr, dth, dp, dut, dur, duth, dup]
```


Python Program: Numerical Integration

To solve the system, we specify the initial conditions and the range of the parameter over which the system is to be solved.

```
initial_conditions = [0, r0, th0, p0, ut0, ur0, uth0, up0]  
tau_span = (0, T)  
tau_eval = np.linspace(*tau_span, P)  
sol = solve_ivp(system, tau_span, initial_conditions, t_eval=tau_eval)
```

Simulation

I have used the ManimCE library to create simulations. It's a Python-based animation engine built for generating precise, programmatic visualisations, well-suited for mathematical and physical concepts.

- Gravitational Lensing Effect
- Photon Sphere
- Circular Motion
- Event Horizon Approach
- Perihelion Precession
- Whirl-Zoom Motion

Gravitational Lensing Effect

- Gravitational lensing is a phenomenon where light from a distant source is bent as it passes near a massive object, such as a star, galaxy, or black hole.
- Depending on the alignment of the source, lens, and observer, gravitational lensing can produce multiple images, apparent magnification, distortions, or even ring-like structures known as Einstein rings.

[Animation Link](#)

Event Horizon Approach

- In Newtonian gravity, a particle speeds up as it falls toward a massive body. The same happens in general relativity at first.
- But as the particle gets very close to the event horizon of a black hole, something interesting happens.
- For a distant observer, the particle appears to slow down and never actually crosses the event horizon.
- Instead, it gets closer and closer, approaching the horizon asymptotically.

[Animation Link](#)

Perihelion Precession

- Perihelion precession is observed in the orbits of planets around a massive body like the Sun.
- In Newtonian gravity, planetary orbits are closed ellipses.
- General relativity predicts that the elliptical orbit slowly rotates over time. This means the point of closest approach to the Sun, known as the perihelion, shifts slightly with each revolution.
- The effect is most famously observed in the orbit of Mercury, whose perihelion advances by a small but measurable amount that could not be fully explained by Newtonian mechanics.
- General relativity accurately accounts for this discrepancy, marking one of its earliest and most successful experimental confirmations.

[Animation Link](#)

Whirl Zoom motion

- Whirl–zoom motion is a unique type of orbital behaviour that occurs in the strong–field regime of general relativity.
- In these trajectories, a particle orbits a central mass along a highly elongated, precessing path.
- As it approaches the periapsis, the particle spends a significant amount of time “whirling” rapidly before “zooming” out to a greater distance, only to fall back in again.
- This repeating pattern creates tight, spiralling loops near the black hole, followed by sweeping arcs at greater distances.

[Animation Link](#)

**Thank you
very much!**